MODEL ARCHITECTURE SPECIFICATION

Predictive Model Plan: Credit Delinquency Risk

Institution: Geldium Financial Services **Author:** Risk Analytics Team **Phase:** Task 2 Conceptual Plan

1. MODEL LOGIC (GENERATED WITH GENAI)

Our chosen modeling strategy is a **Cost-Sensitive Gradient Boosted Decision Tree (LightGBM)** paired with a **Regularized Logistic Regression baseline**. In Task 1's exploratory analysis, linear correlations with delinquency were suppressed ($|r| < 0.05$), indicating that credit defaults stem from complex multi-feature interactions (such as combined debt burden and missed payment streaks) rather than independent linear effects. LightGBM excels at learning non-linear boundary thresholds, directly handles missing values, and natively supports class weight calibration for Geldium's 84:16 imbalanced target distribution.

CORE OBJECTIVE & TOP 5 INPUT FEATURE RANKINGS

The model outputs a calibrated probability of 90-day delinquency: *P(Delinquent_Account = 1 | X)*. The top 5 predictive variables derived from behavioral and financial metrics are:

1. **Recent Payment Recency Score:** Decayed 6-month status weighting recent missed payments heavily.
2. **Credit Utilization Rate:** Measure of revolving credit stress, including over-limit breaches (>100%).
3. **Debt-to-Income (DTI) × Utilization:** Compounded leverage interaction reflecting debt load relative to cash flow.
4. **Standardized Employment Status:** Normalizes consolidated employment stability categories.
5. **Missed Payments (Last 12 Mos):** Baseline annual count of historical credit breaches.

Pipeline Stage	Action Performed	Task 1 EDA Grounding & Rationale
1. Ingestion & Hygiene	Consolidate `EMP`, `employed`, `Employed` into 1 class.	Removes label noise and repairs the 48% fragmented employed segment.
2. Feature Transformation	Map strings `On-time`, `Late`, `Missed` to ordinals (0, 1, 2).	Converts textual sequences into mathematical risk vectors for boosting splits.
3. Feature Engineering	Compute interaction terms & exponential decay streak scores.	Extracts signals from weak standalone bivariate correlations observed in Task 1.
4. In-Fold Imputation	Fit MICE on `Income`; median imputation for `Credit_Score`.	Prevents data leakage across train/validation splits during cross-validation.
5. Probability Calibration	Isotonic calibration with cost-adjusted decision thresholds.	Ensures predicted default probabilities reflect true empirical default frequencies.

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# GenAI-Generated Pipeline Pseudocode (Conceptual Architecture) Input: Raw Customer Record X = [Age, Income, Credit_Score, Utilization, Missed_Payments, Month_1..6, ...] Function Predict_Delinquency_Risk(X, Model_Weights, Threshold = 0.35): X_clean = Harmonize_Employment_Status(X) # Map 'EMP'/'employed' -> 'Employed' X_enc = Ordinal_Encode(Month_1..6) # 'On-time':0, 'Late':1, 'Missed':2 X_feat = Derive_Interaction_Terms(X_clean, X_enc) # Utilization * DTI, Recency_Decay_Score X_imp = Apply_Frozen_Imputation(X_feat) # Impute missing Income/Loan_Balance in-fold P_raw = LightGBM_Tree_Ensemble.Predict_Proba(X_imp) P_calib = Isotonic_Calibrate(P_raw) # Align predicted probabilities with empirical risk Decision = "HIGH RISK / REVIEW" if P_calib >= Threshold else "APPROVED / LOW RISK" Return (Probability = P_calib, Decision = Decision, SHAP_Top_Factors = Explain(X_imp))
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2. JUSTIFICATION FOR MODEL CHOICE

Geldium requires a credit decisioning system that balances high predictive power on rare delinquency events with strict regulatory compliance, transparent auditability, and production stability. While deep neural networks deliver strong predictive capability, their unconstrained black-box nature risks regulatory non-compliance under FCRA and ECOA adverse action notice requirements. Conversely, standard logistic regression proved underpowered in our Task 1 EDA due to the absence of linear correlations. We recommend a **Tree-Based Gradient Boosting Architecture (LightGBM) paired with SHAP (SHapley Additive exPlanations)**. This framework achieves superior discrimination over imbalanced data (16% delinquency), automatically captures compound leverage thresholds (e.g., credit utilization > 100%), and produces exact, mathematically sound adverse action reason codes for every underwriting decision.

Evaluation Vector	Logistic Regression	LightGBM + SHAP (Recommended)	Deep Neural Networks
Non-Linear Capture	Requires manual interaction terms	**High** (natively splits on thresholds)	**Very High** (extracts latent representations)
Interpretability	Direct coefficient weights	**Audit-ready via exact SHAP values**	Low (opaque latent embeddings)
Imbalance Robustness	Prone to under-predicting minority class	**High** (tunable scale_pos_weight)	Moderate (requires complex loss functions)
Deployment Latency	< 5 ms inference	**< 10 ms inference; lightweight**	50–150 ms (heavy compute dependency)

3. EVALUATION STRATEGY & FAIRNESS GOVERNANCE

Because only 16.0% of customer accounts in Geldium's portfolio are delinquent, overall classification accuracy is an uninformative metric (a trivial zero-rule classifier predicting all accounts as non-delinquent achieves 84% accuracy while catching zero credit losses). Our evaluation framework uses risk-calibrated performance metrics alongside fair lending compliance checks.

Evaluation Dimension	Primary Metric Selected	Interpretation & Operational Threshold Criteria
Discrimination Quality	PR-AUC (Precision-Recall) & ROC-AUC	Target PR-AUC > 0.50 (vs 0.16 random base). ROC-AUC target > 0.78 for separation.
Credit Loss Mitigation	Recall at Top Decile (Sensitivity)	Must identify > 70% of all delinquent accounts within highest 20% risk tier.
Fair Lending Equity	Disparate Impact Ratio (DIR)	Approval rate ratio between demographic protected classes must fall within 0.80 – 1.25.
Equal Opportunity	True Positive Rate Parity (TPR)	Difference in false-positive rejections across protected age/employment cohorts < 5%.

ETHICAL SAFEGUARDS & OPERATIONAL GOVERNANCE

- **Exclusion of Sensitive Proxies:** Strictly exclude customer age, location codes, and protected demographic markers from direct model splitting to prevent algorithmic redlining.
- **Adverse Action Transparency:** Every flagged application automatically maps top negative SHAP attributes into human-readable consumer disclosure letters (e.g., "High credit utilization relative to historical earnings").
- **Post-Deployment Drift Monitoring:** Continuous Population Stability Index (PSI) tracking every 30 days. PSI > 0.25 triggers mandatory re-calibration and risk parameter re-validation.